

Lecture 3: Convex sets and convex functions

MA3236 Non-Linear Programming

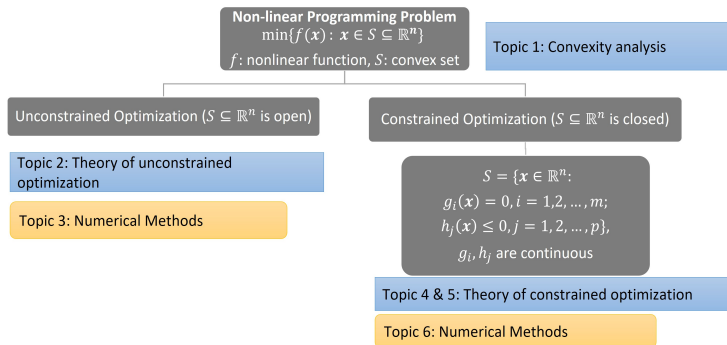
Hu Yukuan

yukuan.hu@nus.edu.sg
S17-08-21

Quick review

- The graphical method provides good intuitions for problems with 1 or 2 variables, and can require computations on dense grids.
- Local and global minimizers are defined topologically, and are the ultimate targets when solving optimization problems.
- Continuous objective plus (i) compact feasible set or (ii) coercivity imply the existence of global minimizer(s). (Sufficient conditions!)

Topics



In lectures 3–5, we will study convex analysis (Topic 1).

- Convex sets and convex functions.
- First-order characterizations of convex functions.
- Second-order characterizations of convex functions.

Why focus on convexity?

Convex objective + Convex feasible set = Convex optimization.

- Wider range of applications than linear optimization.
- Nice optimality properties (in topics 2, 4, 5).
- Many available algorithms (CVXPY).
- Fundamentals of general nonlinear optimization.

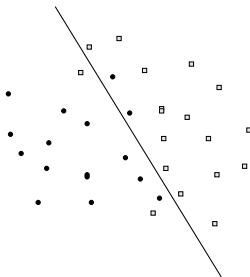


Figure: Support vector machine is a convex optimization.

No gap in convex optimization

Theorem 3.1 (To be proved in the next lecture)

Let C be a nonempty **convex subset** of $\mathbf{R}^n$, and $f : C \rightarrow \mathbf{R}$ is a **convex function**. Consider the convex minimization problem:

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{x} \in C. \end{array}$$

Suppose $\mathbf{x}^* \in C$ is a local minimizer of the problem. Then

- 1 $\mathbf{x}^*$ is a global minimizer.
- 2 if f is strictly convex, then $\mathbf{x}^*$ is the **unique** global minimizer.

Convex sets

Definition 3.2 (Convex set)

A set $C \subseteq \mathbf{R}^n$ is said to be **convex** if for any two points $\mathbf{x}$ and $\mathbf{y}$ in C , the line segment joining $\mathbf{x}$ and $\mathbf{y}$ also lies in C . That is,

$$\mathbf{x}, \mathbf{y} \in C \implies \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C \quad \forall \lambda \in [0, 1].$$

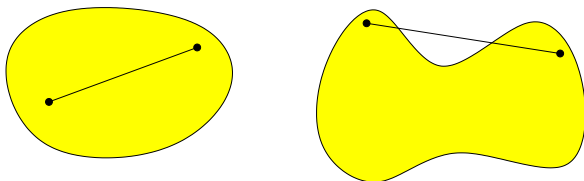


Figure: A convex set and a non-convex set.

Notation: $[\mathbf{x}, \mathbf{y}] := \{\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} : \lambda \in [0, 1]\}$.

To prove a given set C is convex:

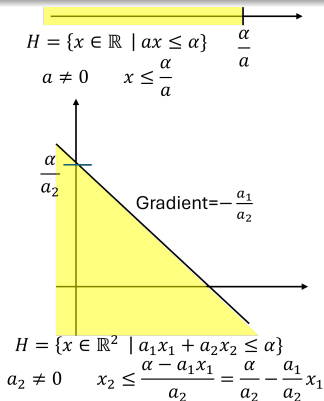
For any two arbitrary points $\mathbf{x}, \mathbf{y}$ in C , and any $\lambda \in [0, 1]$.

Our task is to argue that $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C$.

Example 3.3 (Linear-inequality set)

Given $\mathbf{a} \in \mathbf{R}^n$ and $\alpha \in \mathbf{R}$, the set $H := \{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{a}^T \mathbf{x} \leq \alpha\}$ is convex.

When $\mathbf{a} \neq \mathbf{0}$, it is a half-space.



To prove a given set C is convex:

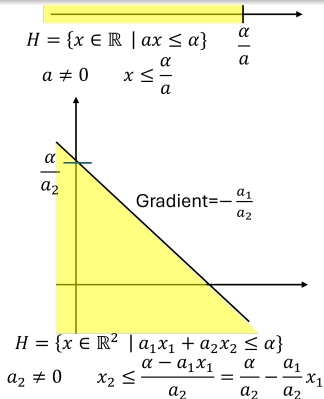
For any two arbitrary points $\mathbf{x}, \mathbf{y}$ in C , and any $\lambda \in [0, 1]$.

Our task is to argue that $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C$.

Example 3.3 (Linear-inequality set)

Given $\mathbf{a} \in \mathbf{R}^n$ and $\alpha \in \mathbf{R}$, the set $H := \{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{a}^T \mathbf{x} \leq \alpha\}$ is convex.

When $\mathbf{a} \neq \mathbf{0}$, it is a half-space.



Proof. Consider $\mathbf{x}, \mathbf{y} \in H$ and $\lambda \in [0, 1]$, and let $\mathbf{w} := \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$. Then

$$\begin{aligned} \mathbf{a}^T \mathbf{w} &= \mathbf{a}^T (\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \\ &= \lambda \mathbf{a}^T \mathbf{x} + (1 - \lambda) \mathbf{a}^T \mathbf{y} \\ &\leq \lambda \alpha + (1 - \lambda) \alpha = \alpha. \end{aligned}$$

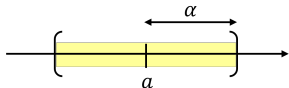
Thus $\mathbf{w} \in H$, so H is convex. □

Example 3.4 (n -ball)

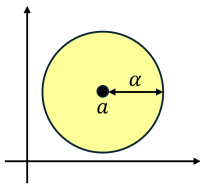
Given $\mathbf{a} \in \mathbf{R}^n, \alpha > 0$. The n -ball

$$\bar{B}(\mathbf{a}, \alpha) := \{\mathbf{x} \in \mathbf{R}^n \mid \|\mathbf{x} - \mathbf{a}\| \leq \alpha\}$$

is a convex set.



$$\begin{aligned}\bar{B}(a, \alpha) &= \{x \in \mathbb{R} \mid |x - a| \leq \alpha\} \\ &= [a - \alpha, a + \alpha]\end{aligned}$$



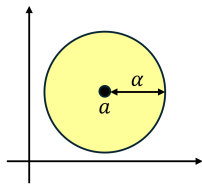
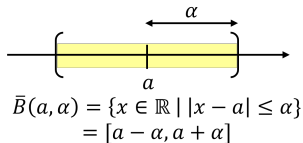
$$\bar{B}(a, \alpha) = \{x \in \mathbb{R}^2 \mid \|x - a\| \leq \alpha\}$$

Example 3.4 (n -ball)

Given $\mathbf{a} \in \mathbf{R}^n, \alpha > 0$. The n -ball

$$\bar{B}(\mathbf{a}, \alpha) := \{\mathbf{x} \in \mathbf{R}^n \mid \|\mathbf{x} - \mathbf{a}\| \leq \alpha\}$$

is a convex set.



$$\bar{B}(a, \alpha) = \{x \in \mathbb{R}^2 \mid \|x - a\| \leq \alpha\}$$

Proof.

Consider $\mathbf{x}, \mathbf{y} \in \bar{B}(\mathbf{a}, \alpha)$ and $\lambda \in [0, 1]$, let $\mathbf{w} := \lambda \mathbf{x} + (1 - \lambda)\mathbf{y}$. Then

$$\begin{aligned} \|\mathbf{w} - \mathbf{a}\| &= \|\lambda(\mathbf{x} - \mathbf{a}) + (1 - \lambda)(\mathbf{y} - \mathbf{a})\| \\ &\leq \lambda\|\mathbf{x} - \mathbf{a}\| + (1 - \lambda)\|\mathbf{y} - \mathbf{a}\| \\ &\leq \alpha. \end{aligned}$$

Thus, $\mathbf{w} \in \bar{B}(\mathbf{a}, \alpha)$, implying $\bar{B}(\mathbf{a}, \alpha)$ is convex. □

Example 3.5

Prove that the set $S := \{\mathbf{x} \in \mathbf{R}^2 \mid x_2 \geq 5, \ x_1^2 - 4x_2 \leq 7\}$ is convex.

Example 3.5

Prove that the set $S := \{\mathbf{x} \in \mathbf{R}^2 \mid x_2 \geq 5, x_1^2 - 4x_2 \leq 7\}$ is convex.

Solution. Let $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$. Then

$$x_2 \geq 5, x_1^2 - 4x_2 \leq 7 \quad \text{and} \quad y_2 \geq 5, y_1^2 - 4y_2 \leq 7.$$

We shall verify that

$$\mathbf{w} := \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} = \begin{pmatrix} \lambda x_1 + (1 - \lambda) y_1 \\ \lambda x_2 + (1 - \lambda) y_2 \end{pmatrix} \in S.$$

Example 3.5

Prove that the set $S := \{\mathbf{x} \in \mathbf{R}^2 \mid x_2 \geq 5, x_1^2 - 4x_2 \leq 7\}$ is convex.

Solution. Let $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$. Then

$$x_2 \geq 5, x_1^2 - 4x_2 \leq 7 \quad \text{and} \quad y_2 \geq 5, y_1^2 - 4y_2 \leq 7.$$

We shall verify that

$$\mathbf{w} := \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} = \begin{pmatrix} \lambda x_1 + (1 - \lambda) y_1 \\ \lambda x_2 + (1 - \lambda) y_2 \end{pmatrix} \in S.$$

(i) $x_2 \geq 5, y_2 \geq 5$ and $\lambda \geq 0, (1 - \lambda) \geq 0$

$$\implies w_2 = \lambda x_2 + (1 - \lambda) y_2 \geq 5\lambda + 5(1 - \lambda) = 5.$$

Example 3.5 (cont'd)

(ii) Using $2x_1y_1 \leq x_1^2 + y_1^2$,

$$w_1^2 - 4w_2 = (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2)$$

Therefore, $\mathbf{w} = \lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.

Example 3.5 (cont'd)

(ii) Using $2x_1y_1 \leq x_1^2 + y_1^2$,

$$\begin{aligned}w_1^2 - 4w_2 &= (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\&= \lambda^2 x_1^2 + 2\lambda(1 - \lambda)x_1y_1 + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2)\end{aligned}$$

Therefore, $\mathbf{w} = \lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.

Example 3.5 (cont'd)

(ii) Using $2x_1y_1 \leq x_1^2 + y_1^2$,

$$\begin{aligned}w_1^2 - 4w_2 &= (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\&= \lambda^2 x_1^2 + 2\lambda(1 - \lambda)x_1y_1 + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\&\leq \lambda(x_1^2 - 4x_2) + (1 - \lambda)(y_1^2 - 4y_2)\end{aligned}$$

Therefore, $\mathbf{w} = \lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.

Example 3.5 (cont'd)

(ii) Using $2x_1y_1 \leq x_1^2 + y_1^2$,

$$\begin{aligned}w_1^2 - 4w_2 &= (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\&= \lambda^2 x_1^2 + 2\lambda(1 - \lambda)x_1y_1 + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\&\leq \lambda(x_1^2 - 4x_2) + (1 - \lambda)(y_1^2 - 4y_2) \\&\leq 7\lambda + 7(1 - \lambda) = 7.\end{aligned}$$

Therefore, $\mathbf{w} = \lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.

Question. Is the circle defined by $\{\mathbf{x} \in \mathbf{R}^2 | x_1^2 + x_2^2 = 1\}$ a convex set?

Proposition 3.6

Let m be a positive integer. If $C_1, C_2, \dots, C_m$ are convex sets in $\mathbf{R}^n$, then $C := \cap_{i=1}^m C_i$ is also convex.

Intersection of convex sets

Proposition 3.6

Let m be a positive integer. If $C_1, C_2, \dots, C_m$ are convex sets in $\mathbf{R}^n$, then $C := \cap_{i=1}^m C_i$ is also convex.

Proof. Let $\mathbf{x}, \mathbf{y} \in \cap_{i=1}^m C_i$ and $\lambda \in [0, 1]$. Then $\mathbf{x}, \mathbf{y} \in C_i \ \forall i = 1, 2, \dots, m$.

Since each C_i is convex, $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C_i \ \forall i = 1, 2, \dots, m$.

Thus, $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in \cap_{i=1}^m C_i$. Therefore $\cap_{i=1}^m C_i$ is convex. □

Intersection of convex sets

Proposition 3.6

Let m be a positive integer. If $C_1, C_2, \dots, C_m$ are convex sets in $\mathbf{R}^n$, then $C := \cap_{i=1}^m C_i$ is also convex.

Proof. Let $\mathbf{x}, \mathbf{y} \in \cap_{i=1}^m C_i$ and $\lambda \in [0, 1]$. Then $\mathbf{x}, \mathbf{y} \in C_i \ \forall i = 1, 2, \dots, m$.

Since each C_i is convex, $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in C_i \ \forall i = 1, 2, \dots, m$.

Thus, $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in \cap_{i=1}^m C_i$. Therefore $\cap_{i=1}^m C_i$ is convex. □

Question: how about the case of infinite convex sets?

Example 3.7 (Polyhedron or polyhedral set)

For positive integers m, n , let $\mathbf{A} \in \mathbf{R}^{m \times n}$ and $\mathbf{b} \in \mathbf{R}^m$. The polyhedron $\{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{Ax} \leq \mathbf{b}\}$ is convex.

Example 3.7 (Polyhedron or polyhedral set)

For positive integers m, n , let $\mathbf{A} \in \mathbf{R}^{m \times n}$ and $\mathbf{b} \in \mathbf{R}^m$. The polyhedron $\{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{Ax} \leq \mathbf{b}\}$ is convex.

Proof. Write

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix}, \quad \mathbf{a}_i \in \mathbf{R}^n, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Example 3.7 (Polyhedron or polyhedral set)

For positive integers m, n , let $\mathbf{A} \in \mathbf{R}^{m \times n}$ and $\mathbf{b} \in \mathbf{R}^m$. The polyhedron $\{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{Ax} \leq \mathbf{b}\}$ is convex.

Proof. Write

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix}, \quad \mathbf{a}_i \in \mathbf{R}^n, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{aligned} \{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{Ax} \leq \mathbf{b}\} &= \{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{a}_1^T \mathbf{x} \leq b_1, \dots, \mathbf{a}_m^T \mathbf{x} \leq b_m\} \\ &= \bigcap_{i=1}^m \{\mathbf{x} \in \mathbf{R}^n \mid \mathbf{a}_i^T \mathbf{x} \leq b_i\}. \end{aligned}$$

By Example 3.3 and Proposition 3.6.



Question. Suppose $C_1, C_2, \dots, C_m$ are convex sets in $\mathbf{R}^n$. Is the set $C := \bigcup_{i=1}^m C_i$ convex?

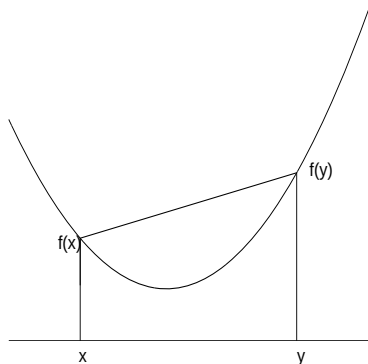
Convex functions

Let $C \subseteq \mathbf{R}^n$ be a **convex** set. Consider a function $f : C \rightarrow \mathbf{R}$.

The function f is said to be **convex** if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in C, \lambda \in [0, 1].$$

- the chord joining $(\mathbf{x}, f(\mathbf{x}))$ and $(\mathbf{y}, f(\mathbf{y}))$ lies **above** the graph of f along $[\mathbf{x}, \mathbf{y}]$.

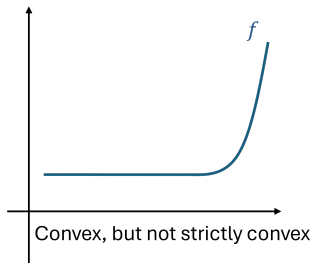
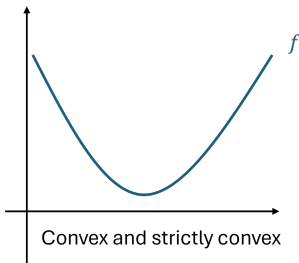


Definition 3.8 (Strictly convex function)

Let $C \subseteq \mathbf{R}^n$ be a **convex** set. Consider a function $f : C \rightarrow \mathbf{R}$. The function f is said to be **strictly convex** if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) < \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}),$$

for all distinct $\mathbf{x}, \mathbf{y} \in C, \lambda \in (0, 1)$.



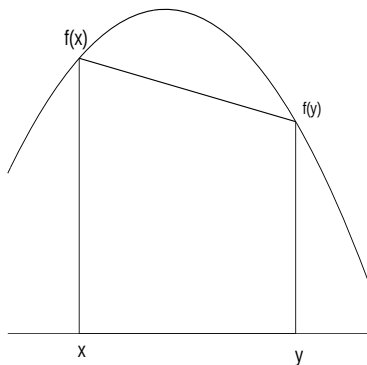
Concave functions

Let $C \subseteq \mathbf{R}^n$ be a **convex** set. Consider a function $f : C \rightarrow \mathbf{R}$.

The function f is said to be **concave** if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \geq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}), \quad \forall \mathbf{x}, \mathbf{y} \in C, \lambda \in [0, 1].$$

- the chord joining $(\mathbf{x}, f(\mathbf{x}))$ and $(\mathbf{y}, f(\mathbf{y}))$ lies **below** the graph of f along $[\mathbf{x}, \mathbf{y}]$.

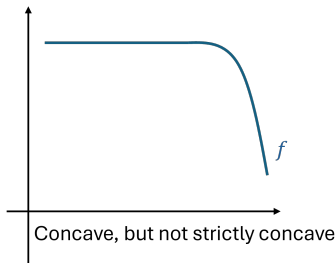
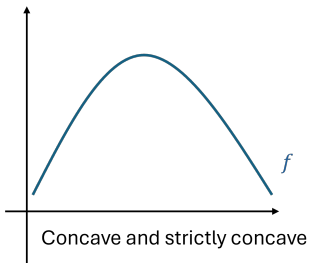


Definition 3.9 (Strictly concave function)

Let $C \subseteq \mathbf{R}^n$ be a **convex** set. Consider a function $f : C \rightarrow \mathbf{R}$. The function f is said to be **strictly concave** if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) > \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}),$$

for all distinct $\mathbf{x}, \mathbf{y} \in C, \lambda \in (0, 1)$.

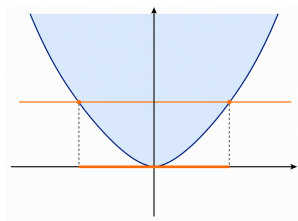


Epigraphs and sublevel sets

Let $C \subseteq \mathbf{R}^n$ be convex and let $f : C \rightarrow \mathbf{R}$.

Epigraph

$$\text{epi } f := \{(\mathbf{x}, t) \in C \times \mathbf{R} : f(\mathbf{x}) \leq t\}.$$



Blue region: $\text{epi } f$;
orange line: $t = \alpha$;
orange interval: $L_\alpha(f)$.

For $\alpha \in \mathbf{R}$, the α -sublevel set is

$$L_\alpha(f) := \{\mathbf{x} \in C : f(\mathbf{x}) \leq \alpha\}.$$

It is a horizontal slice of the epigraph:

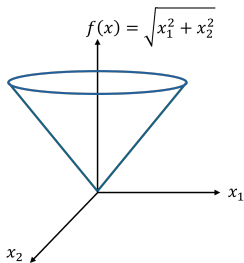
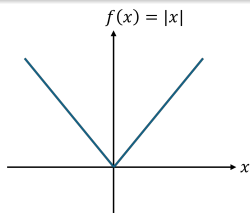
$$L_\alpha(f) \times \{\alpha\} = \text{epi } f \cap (\mathbf{R}^n \times \{\alpha\}).$$

$$\begin{aligned} f \text{ is convex} &\iff \text{epi } f \text{ is convex} \\ &\implies L_\alpha(f) \text{ is convex } (\forall \alpha) \end{aligned}$$

Question. How about the reverse direction?

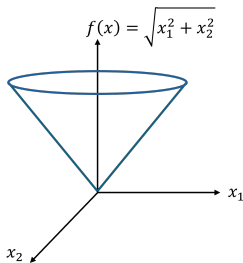
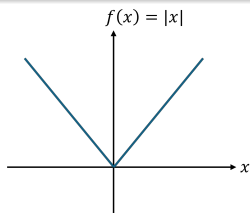
Example 3.10

Let $f : \mathbf{R}^2 \rightarrow \mathbf{R}$, where $f(\mathbf{x}) := \|\mathbf{x}\| = (x_1^2 + x_2^2)^{1/2}$. Show that f is convex in $\mathbf{R}^2$.



Example 3.10

Let $f : \mathbf{R}^2 \rightarrow \mathbf{R}$, where $f(\mathbf{x}) := \|\mathbf{x}\| = (x_1^2 + x_2^2)^{1/2}$. Show that f is convex in $\mathbf{R}^2$.



Solution. Let $\mathbf{x}, \mathbf{y} \in \mathbf{R}^2$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned} & f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \\ &= \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\| \\ &\leq \|\lambda \mathbf{x}\| + \|(1 - \lambda) \mathbf{y}\| \\ &= \lambda \|\mathbf{x}\| + (1 - \lambda) \|\mathbf{y}\| \\ &= \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) \end{aligned}$$

Therefore, f is a convex function.

Example 3.11

Let $\mathbf{a} \in \mathbf{R}^n$ and $\alpha \in \mathbf{R}$. Consider $f : \mathbf{R}^n \rightarrow \mathbf{R}$, $f(\mathbf{x}) := \mathbf{a}^T \mathbf{x} + \alpha$.

f is convex, but not strictly convex.

Note that f is also a concave function.

Example 3.11

Let $\mathbf{a} \in \mathbf{R}^n$ and $\alpha \in \mathbf{R}$. Consider $f : \mathbf{R}^n \rightarrow \mathbf{R}$, $f(\mathbf{x}) := \mathbf{a}^T \mathbf{x} + \alpha$.

f is convex, but not strictly convex.

Note that f is also a concave function.

Question: If a function is both convex and concave, must it be affine?

Example 3.12

Prove that the function $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ defined by $f(\mathbf{x}) := x_1^2 - 4x_2$ is convex.

Solution. Consider $\mathbf{x}, \mathbf{y} \in \mathbf{R}^2$ and $\lambda \in [0, 1]$. Let $\mathbf{w} := \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. Then using $2x_1y_1 \leq x_1^2 + y_1^2$,

$$\begin{aligned} f(\mathbf{w}) &= (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &= \lambda^2 x_1^2 + 2\lambda(1 - \lambda)x_1y_1 + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &\leq \lambda^2 x_1^2 + \lambda(1 - \lambda)(x_1^2 + y_1^2) + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &= \lambda(x_1^2 - 4x_2) + (1 - \lambda)(y_1^2 - 4y_2) \\ &= \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}). \end{aligned}$$

Example 3.12

Prove that the function $f : \mathbf{R}^2 \rightarrow \mathbf{R}$ defined by $f(\mathbf{x}) := x_1^2 - 4x_2$ is convex.

Solution. Consider $\mathbf{x}, \mathbf{y} \in \mathbf{R}^2$ and $\lambda \in [0, 1]$. Let $\mathbf{w} := \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. Then using $2x_1y_1 \leq x_1^2 + y_1^2$,

$$\begin{aligned} f(\mathbf{w}) &= (\lambda x_1 + (1 - \lambda)y_1)^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &= \lambda^2 x_1^2 + 2\lambda(1 - \lambda)x_1y_1 + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &\leq \lambda^2 x_1^2 + \lambda(1 - \lambda)(x_1^2 + y_1^2) + (1 - \lambda)^2 y_1^2 - 4(\lambda x_2 + (1 - \lambda)y_2) \\ &= \lambda(x_1^2 - 4x_2) + (1 - \lambda)(y_1^2 - 4y_2) \\ &= \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}). \end{aligned}$$

Question 1 (revisit Example 3.5): Is $\{\mathbf{x} : x_2 \geq 5, x_1^2 - 4x_2 \leq 7\}$ convex?

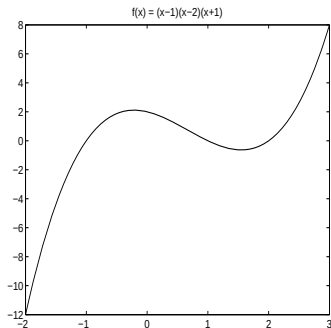
Question 2. Is f strictly convex?

A function which is neither convex nor concave

Define $f : [-2, 3] \rightarrow \mathbf{R}$ by $f(x) := (x-1)(x-2)(x+1)$. Then

$$f(-1) = 0 > \frac{f(-2) + f(0)}{2} = -5, \quad f(1) = 0 < \frac{f(0) + f(2)}{2} = 1.$$

Thus f is neither convex nor concave on $[-2, 3]$.



Convex functions: useful properties

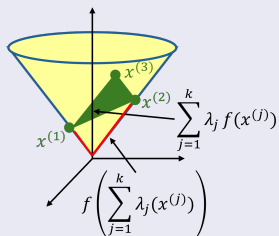
Proposition 3.13 (Jensen's inequality)

Let $k \geq 2$ be an integer, let $C \subseteq \mathbf{R}^n$ be convex, let $f : C \rightarrow \mathbf{R}$ be convex, and let $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} \in C$. Choose weights $\lambda_1, \dots, \lambda_k \geq 0$ satisfying $\sum_{j=1}^k \lambda_j = 1$, and define

$$\mathbf{x} := \sum_{j=1}^k \lambda_j \mathbf{x}^{(j)} \in C.$$

Then

$$f(\mathbf{x}) \leq \sum_{j=1}^k \lambda_j f(\mathbf{x}^{(j)}).$$

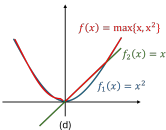
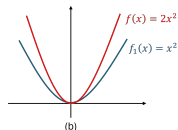
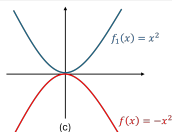
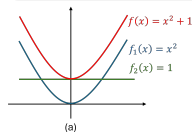


Convex functions: useful properties (cont'd)

Proposition 3.14 (Operations preserving convexity)

Let $C \subseteq \mathbf{R}^n$ be convex. If $f_1, f_2 : C \rightarrow \mathbf{R}$ are convex, then

- (a) $f_1 + f_2$ is a convex function on C .
- (b) αf_1 is a convex function on C for $\alpha \geq 0$.
- (c) αf_1 is a concave function on C for $\alpha < 0$.
- (d) The pointwise maximum $g(\mathbf{x}) := \max\{f_1(\mathbf{x}), f_2(\mathbf{x})\}$ is convex on C .

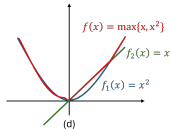
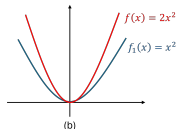
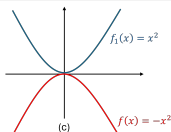
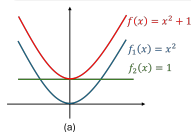


Convex functions: useful properties (cont'd)

Proposition 3.14 (Operations preserving convexity)

Let $C \subseteq \mathbf{R}^n$ be convex. If $f_1, f_2 : C \rightarrow \mathbf{R}$ are convex, then

- (a) $f_1 + f_2$ is a convex function on C .
- (b) αf_1 is a convex function on C for $\alpha \geq 0$.
- (c) αf_1 is a concave function on C for $\alpha < 0$.
- (d) The pointwise maximum $g(\mathbf{x}) := \max\{f_1(\mathbf{x}), f_2(\mathbf{x})\}$ is convex on C .



Question. How about pointwise minimum?

Corollary 3.15

Let k be a positive integer, let $C \subseteq \mathbf{R}^n$ be convex, and let $f_1, \dots, f_k : C \rightarrow \mathbf{R}$ be convex. For coefficients $\alpha_1, \dots, \alpha_k \geq 0$, the function

$$f(\mathbf{x}) := \sum_{j=1}^k \alpha_j f_j(\mathbf{x})$$

is also convex on C . Moreover, if f_j is strictly convex on C and $\alpha_j > 0$ for some j , then f is strictly convex on C .

Composite functions

Proposition 3.16

Let $C \subseteq \mathbf{R}^n$ be convex, let $h : C \rightarrow \mathbf{R}$ be convex, and let $I \subseteq \mathbf{R}$ be an interval containing $h(C)$. If $g : I \rightarrow \mathbf{R}$ is convex and nondecreasing, then $f := g \circ h$ is convex on C .

Composite functions

Proposition 3.16

Let $C \subseteq \mathbf{R}^n$ be convex, let $h : C \rightarrow \mathbf{R}$ be convex, and let $I \subseteq \mathbf{R}$ be an interval containing $h(C)$. If $g : I \rightarrow \mathbf{R}$ is convex and nondecreasing, then $f := g \circ h$ is convex on C .

Proof. For $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$,

$$g(h(\lambda\mathbf{x} + (1-\lambda)\mathbf{y})) \leq g(\lambda h(\mathbf{x}) + (1-\lambda)h(\mathbf{y})) \leq \lambda g(h(\mathbf{x})) + (1-\lambda)g(h(\mathbf{y})).$$

Composite functions

Proposition 3.16

Let $C \subseteq \mathbf{R}^n$ be convex, let $h : C \rightarrow \mathbf{R}$ be convex, and let $I \subseteq \mathbf{R}$ be an interval containing $h(C)$. If $g : I \rightarrow \mathbf{R}$ is convex and nondecreasing, then $f := g \circ h$ is convex on C .

Proof. For $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$,

$$g(h(\lambda\mathbf{x} + (1-\lambda)\mathbf{y})) \leq g(\lambda h(\mathbf{x}) + (1-\lambda)h(\mathbf{y})) \leq \lambda g(h(\mathbf{x})) + (1-\lambda)g(h(\mathbf{y})).$$

Question. Is the above result true if g is a convex function but it is not nondecreasing?

Example 3.17

Suppose $f : \mathbf{R} \rightarrow \mathbf{R}$ is given by $f(x) := x^2 + x$. Show that the following functions are convex:

(i) $\exp(x^2 + x)$, $x \in \mathbf{R}$

(ii) $(x^2 + x)^2$, $x \in [0, \infty)$.

Proposition 3.18

Let $C \subseteq \mathbf{R}^n$ be convex, let $h : C \rightarrow \mathbf{R}$ be convex, and let $I \subseteq \mathbf{R}$ be an interval containing $h(C)$. If $g : I \rightarrow \mathbf{R}$ is concave and nonincreasing, then $f := g \circ h$ is concave on C .

Proof. For $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$,

$$g(h(\lambda\mathbf{x} + (1-\lambda)\mathbf{y})) \geq g(\lambda h(\mathbf{x}) + (1-\lambda)h(\mathbf{y})) \geq \lambda g(h(\mathbf{x})) + (1-\lambda)g(h(\mathbf{y})).$$

Take-home messages

- Convex optimization is special yet useful, possessing nice theoretical properties (shown in later lectures).
- Convex sets contain the line segment joining any two of their points; convex functions satisfy “convex inequalities”.
- Operations that preserve convexity (sets / functions).
- In terms of convexity, sets and functions are not far from each other.
- Question: more analytical ways to verify convexity?

Historical credits

Convex geometry

- Karl Hermann Brunn (1862-1939).
- Hermann Minkowski (1864-1909).

Convex functions and early duality

- Otto Stolz (1842-1905).
- Johan Jensen (1859-1925).
- Ludwig Otto Hölder (1859-1937).
- Moritz Werner Fenchel (1905-1988).

Modern convex analysis

- Jean-Jacques Moreau (1923-2014).
- Ralph Tyrrell Rockafellar (1935-). [Latest single-authored paper in 2025.](#)

The list is definitely not comprehensive...